Indian Institute of Information Technology Vadodara
MA441: Mathematics I (Discrete Mathematics)
Tutorial 1

1. What is $\phi \cup \{\phi\}$?
2. Let A be the set of all digits in your student id. Write down A. Is $\phi \subseteq A$? Is $5 \subseteq A$? Give justification.
3. Let A denote the set of all straight lines in the plane $\mathbb{R}^2$. Does S, T, or both belong to A? $S = \{(x, y) : y = 2x + 7\}, T = \{2, 4, 6, 9\}$.
4. In a class of 100 students, 35 like science and 45 like math. 10 like both. How many like either of them and how many like neither?
5. Find total number of natural numbers which either divides 1800 or 2460.
6. Find $\cup_{i=1}^{n} A_i$ and $\cap_{i=1}^{n} A_i$ for the following:
   (a) $A_i = \{0, i\}$
   (b) $A_i = \{\cdots, -2, -1, 0, 1, 2, \cdots, i\}$
   (c) $A_i = \{i, i + 1, i + 2, \cdots\}$
7. Is associative law: $(A - B) - C = A - (B - C)$ true for any sets A, B, C?
8. For finite sets A, B, C, find formulae for $|A - B|, |A \oplus B|, |(A - B) - C|, |A \cap B \cap C|$.
9. Under what condition following is true?
   $(A - B) \cup (A - C) = A$, for the sets A, B, C.
10. Given two sets A, B, what can you say about $P(A \cup B), P(A \cap B)$ in terms of $P(A), P(B)$?

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Tutorial 2

1. For each of the sequence of integers given below, give a simple formula or rule that gives any term of the sequence which start with the list.
   (a) $15, 8, 1, -6, -13, -20, -27, \ldots$
   (b) $3, 5, 8, 12, 17, 23, 30, \ldots$
   (c) $2, 3, 7, 25, 121, 721, 5041, 40321, \ldots$
2. Find the value of following sums
   (a) $\sum_{j=0}^{8} 3.2^j$
   (b) $\sum_{j=0}^{8} 3.(-2)^j$
   (c) $\sum_{i=0}^{2} \sum_{j=0}^{3} i^2 j^3$
   (d) $\sum_{i=0}^{2} \sum_{j=0}^{3} 1$
3. Determine whether following functions $f : \mathbb{Z} \to \mathbb{Z}$ are injective/one-to-one and or onto?
   (a) $f(n) = n^3$
   (b) $f(n) = n^2 + 1$
   (c) $f(n) = \lceil n/2 \rceil$, where $\lceil x \rceil$ is the smallest integer $\geq x$.
4. Determine whether following functions $f : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ are injective/one-to-one and or surjective/onto?
   (a) $f(m, n) = 2m - n$
   (b) $f(m, n) = m^2 - n^2$
   (c) $f(m, n) = |m| - |n|$
5. Determine whether each of the following functions $f : \mathbb{R} \to \mathbb{R}$ are bijective?
   (a) $f(x) = x^2 + 1$
   (b) $f(x) = x^3$
   (c) $f(x) = \frac{x^2+1}{x^2+2}$.
6. Find an example of functions f and g such that $f \circ g$ is a bijection, but g is not onto and f is not one-to-one.
7. Show that $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = ax + b$, where a, b are real constants with $a \neq 0$ is invertible. Find the inverse of f.
8. Suppose that $g : A \to B$ and $f : B \to C$ are surjective functions. Is $f \circ g$ also surjective?
9. Suppose that f is a function from A to B. We define the function S_f from $P(A)$ to $P(B)$ by the rule $S_f(X) = f(X)$ for each subset X of A. Choose finite sets A, B and define an injective function $f : A \to B$. Describe S_f and check injectivity of S_f. If f is injective then show that S_f is injective. Can you prove similar statement for surjectivity of S_f? What about converse?
10. Find a bijective map $f : [1, 2] \to [3, 6]$. Generalize it for any two intervals $[a,\ b], [c,\ d]$.

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Tutorial 3

1. Find a bijective map $f : [1,2] \to [3,6]$ thereby proving that cardinality of $[1,2]$ and $[3,6]$ are same. Generalize it.

2. Show that the set of all rational numbers is countable.

3. Determine whether each of the following sets is countable or uncountable(not countable).

   (a) $B = \{(x,y) | x \in \mathbb{N}, y \in \mathbb{Z} - \{0\}\}$.

   (b) $C = \mathbb{R} \setminus \mathbb{Q}$.

   (c) $A =$ set of all complex numbers.

4. Show that the set A, of all real numbers that are solutions of quadratic equations $ax^2+bx+c = 0$, for some integers a, b, c, is countable.

5. Find a bijective map $f : P(\mathbb{N}) \to (0,1)$ thereby proving that

6. Prove that $|\mathbb{R}| = |(0,1)|$ and $|\mathbb{R}| = |\mathbb{R}^2|$.

7. Show that the set of all binary strings is countable, thereby proving the set of all programs is countable.

8. Consider a function $f(m,n) = m + \frac{(m+n)(m+n+1)}{2}$ from $\mathbb{N}_{\geq 0} \times \mathbb{N}_{\geq 0}$ to $\mathbb{N}_{\geq 0}$. Is f bijective?

   (HILBERT'S GRAND HOTEL) We now describe a paradox that shows that something impossible with finite sets may be possible with infinite sets. The famous mathematician David Hilbert invented the notion of the Grand Hotel, which has a countably infinite number of rooms, each occupied by a guest. When a new guest arrives at a hotel with a finite number of rooms, and all rooms are occupied, this guest cannot be accommodated without evicting a current guest. However, we can always accommodate a new guest at the Grand Hotel, even when all rooms are already occupied. How can we accommodate a new guest arriving at the fully occupied Grand Hotel without removing any of the current guests?

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Tutorial 4

1. Which of these sentences are propositions? What are the truth values of those that are propositions?
   i) Answer this question.
   ii) $x = x$.
   iii) This statement is true.

2. Write the propositions using p and q and logical connectives. p- Mahesh chooses MA401 as science elective; q-Mahesh likes MA102.
   Either Mahesh does not like MA102 or he does not choose MA401.

3. Suppose there is an island of knights and knaves, where knights always tell the truth and knaves always lie. You encounter two people, A and B. Determine, if possible, what A and B are if they address you in the ways described. If you cannot determine what these two people are, can you draw any conclusions?
   Both A and B say "I am a knight."

4. Express the statement "If a person is female and is a parent, then this person is someone's mother" as a logical expression involving predicates, quantifiers with a domain consisting of all people, and logical connectives.

5. Express the negation of the statement $\forall x \exists y(xy = 1)$ so that no negation precedes a quantifier.

6. Use rules of inference to show that if $\forall x(P(x) \Rightarrow (Q(x) \wedge S(x)))$ and $\forall x(P(x) \wedge R(x))$ are true then $\forall x(R(x) \wedge S(x))$ is true.

7. For following set of premises, what relevant conclusion or conclusions can be drawn? Explain the rules of inference used to obtain each conclusion from the premises
   i) All lobstermen set at least a dozen traps. Hamilton is a lobsterman. Therefore, Hamilton sets at least a dozen traps.
   ii) All foods that are healthy to eat do not taste good.
   Tofu is healthy to eat.
   You only eat what tastes good.
   You do not eat tofu.
   Cheeseburgers are not healthy to eat.

8. What rules of inference are used in this argument? "No man is an island. Manhattan is an island. Therefore, Manhattan is not a man."

9. Use resolution to show that the hypotheses "It is not raining or Geeta has her umbrella," "Geeta does not have her umbrella or she does not get wet," and "It is raining or Geeta does not get wet" imply that "Geeta does not get wet."

10. What is wrong with this argument? Let $H(x)$ be "x is happy." Given the premise $\exists x H(x)$, we conclude that $H(Lola)$. Therefore, Lola is happy.

11. Determine whether this argument is valid.
    If Superman were able and willing to prevent evil, he would do so. If Superman were unable to prevent evil, he would be impotent; if he were unwilling to prevent evil, he would be malevolent. Superman does not prevent evil. If Superman exists, he is neither impotent nor malevolent. Therefore, Superman does not exist.

12. Find DNF, CNF, PDNF, PCNF of the following formula: $P \Rightarrow (P \wedge (Q \Rightarrow P))$

13. The nth statement in a list of 100 statements is "Exactly n of the statements in this list are false." What conclusions can you draw from these statements?

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Tutorial 5

1. Use a proof by contradiction to prove that the sum of an irrational number and a rational number is irrational.

2. Prove that if x is irrational, then $1/x$ is irrational. What method did you use?

3. Prove that if n is an integer and $3n + 2$ is even, then n is even using
   a) a proof by contraposition.
   b) a proof by contradiction.

4. Use a direct proof to show that every odd integer is the difference of two squares.

5. Prove that if n is an integer, these four statements are equivalent: (i) n is even, (ii) $n + 1$ is odd, (iii) $3n + 1$ is odd, (iv) $3n$ is even.

6. Find integers a, b, c, m which do not satisfy following statement:
   If $ac \equiv bc$ mod(m) with $m \geq 2$, then $a \equiv b$ mod(m).

7. Find a formula for gcd of $2^m - 1$ and $2^n - 1$ for $m, n \geq 1$.

8. If p is prime, then find gcd$(p - 1, p + 1)$.

9. Show that $2^{340} \equiv 1$mod(31).

10. Find the last digit of 333^{555}.

Indian Institute of Information Technology Vadodara
MA 441: Mathematics I (Discrete Mathematics)
Tutorial 6

1. Find the remainder when 29! is divided by 31.
2. Find the last two digits of 333^{555}.
3. State and prove divisibility test of 7. Can you design divisibility test of 13?
4. What is the remainder obtained when $2^{70} + 3^{70}$ is divided by 13?
5. Find the multiplicative inverse of each non-zero element of $\mathbb{Z}_{11}$ to verify $\mathbb{Z}_{11}$ is a field.
6. Find all integers x such that $2x \equiv 3 \bmod 5, 3x \equiv 4 \bmod 7, x \equiv 5 \bmod 11$.
7. Use Fermat's little theorem to compute 5^{2023} mod 7, 5^{2023} mod 11, and 5^{2023}mod 13. Use Chinese remainder theorem to find 5^{2023} mod 1001.
8. Prove that if n is a positive integer such that the sum of the positive divisors of n is n + 1, then n is prime. Is converse true?
9. Show that if $2^n - 1$ is a prime then n is also a prime.
10. Is the converse of Fermat's little theorem true?
11. Using Fermat's little theorem show that if n is a positive integer, then 42 divides $n^7 - n$.

Indian Institute of Information Technology Vadodara
MS 441: Mathematics I (Discrete Mathematics)
Tutorial 7

Notation: Let f_n denote the n^{th} Fibonacci number.

1. Guess and prove the formula for $1.1! + 2.2! + 3.3! + \cdots + n.n!$. Use Mathematical induction.

2. Prove, using induction, that a set with n elements has $n(n-1)(n-2)/6$ subsets containing exactly three elements whenever n is an integer greater than or equal to 3.

3. Show that n lines separate the plane into $(n^2+n+2)/2$ regions if no two of these lines are parallel and no three pass through a common point.

4. Suppose there are n people in a group, each aware of a scandal no one else in the group knows about. These people communicate by telephone; when two people in the group talk, they share information about all scandals each knows about. For example, on the first call, two people share information, so by the end of the call, each of these people knows about two scandals. The gossip problem asks for $G(n)$, the minimum number of telephone calls that are needed for all n people to learn about all the scandals. Find $G(2), G(3)$ and $G(4)$. Prove that $G(n) = 2n-4, n \geq 4$.

5. Show that $\displaystyle\sum_{\{a_1,a_2,\ldots,a_k\}\subseteq\{1,2,\ldots,n\}} \frac{1}{a_1.a_2.\ldots.a_k} = n$. Here the sum is over all nonempty subsets of the set of the n smallest positive integers.

6. Let $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. Show that $A^n = \begin{bmatrix} f_{n+1} & f_n \\ f_n & f_{n-1} \end{bmatrix}$ when n is a positive integer.

7. Prove that $f_1 + f_3 + \cdots + f_{2n-1} = f_{2n}$ when n is a positive integer.

8. A partition of a positive integer n is a way to write n as a sum of positive integers where the order of terms in the sum does not matter. For instance, 7= 3 + 2 + 1 + 1 is a partition of 7. Let P_m equal the number of different partitions of m, and let $P_{m,n}$ be the number of different ways to express m as the sum of positive integers not exceeding n. Show that
a) $P_{m,m} = P_m$;
b) find recursive definition of $P_{m,n}$;
c) Use it to find P_6.

Indian Institute of Information Technology Vadodara
MS 441: Mathematics I (Discrete Mathematics)
Tutorial 8

1. How many functions are there from the set $\{1, 2, \cdots, n\}$, where n is a positive integer, to the set $\{0, 1\}$?
2. How many positive integers between 1000 and 9999 inclusive are divisible by 9?
3. How many bit strings of length seven either begin with two 0s or end with three 1s?
4. How many bit strings of length seven either begin with two 0s and end with three 1s?
5. A key in the Vigenère cryptosystem is a string of English letters, where the case of the letters does not matter. How many different keys for this cryptosystem are there with three, four, five, or six letters?
6. Show that if there are 30 students in a class, then at least two have last names that begin with the same letter.
7. A bowl contains 10 red balls and 10 blue balls. A woman selects balls at random without looking at them. How many balls must she select to be sure of having at least three balls of the same color?
8. Find an increasing subsequence of maximal length and a decreasing subsequence of maximal length in the sequence 22, 5, 7, 2, 23, 10, 15, 21, 3, 17.
9. Describe an algorithm in pseudocode for producing the largest increasing or decreasing subsequence of a sequence of distinct integers.
10. Show that if there are 101 people of different heights standing in a line, it is possible to find 11 people in the order they are standing in the line with heights that are either increasing or decreasing.
11. Let $(x_i, y_i, z_i), i = 1, 2, 3, 4, 5, 6, 7, 8, 9$, be a set of nine distinct points with integer coordinates in xyz space. Show that the midpoint of at least one pair of these points has integer coordinates.

Indian Institute of Information Technology Vadodara
MA 101: Introduction to Discrete Mathematics
Tutorial 9

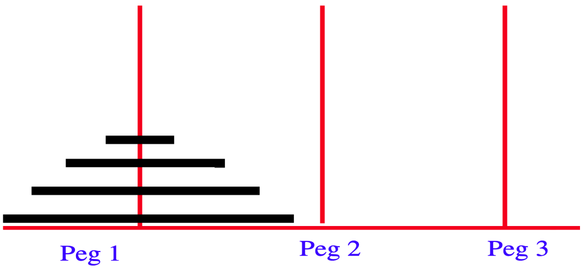
1. You have access to 1×1 tiles which come in 2 different colours and 1×2 tiles which come in 3 different colours. Figure out how many different $1 \times n$ path designs one can make out of these tiles. Find out recurrence relation and closed formula for it.
2. Solve the recurrence relation $a_n = 2a_{n-1} - a_{n-2}$ with initial term $a_0 = 1, a_1 = 2$.
3. Solve the recurrence relation $a_n = a_{n-1}^2 + n$ with initial term $a_0 = 4$.
4. Write a recurrence relation and initial conditions for the numbers s_n of n-bit strings having no two consecutive zeros. Solve it.
5. Find a recurrence relation for the number of ternary strings of length n that contain two consecutive 0s. Solve it.
6. Find the solution of the recurrence relation $a_n = a_{n-1} + 6a_{n-2}$ for $n \geq 2$, $a_0 = 3, a_1 = 6$. Also
7. Find the solution of the recurrence relation $a_n = 5a_{n-2} - 4a_{n-4}$ with $a_0 = 3, a_1 = 2, a_2 = 6$, and $a_3 = 8$.
8. Find the solution of the recurrence relation $a_n = 2a_{n-1} + 2n^2$ with initial condition $a_1 = 4$.
9. Find all solutions of the recurrence relation $a_n = 7a_{n-1} - 16a_{n-2} + 12a_{n-3} + n4^n$ with $a_0 = -2, a_1 = 0$, and $a_2 = 5$.

Indian Institute of Information Technology Vadodara
MS441: Discrete Mathematics
Tutorial 10

1. How many non-negative integer solutions are there to the equation $x_1+x_2+\cdots+x_5 = 21$ with $x_1, x_2 \geq 1, x_3 = 3, x_4 \geq 2, x_5 \geq 0$?

2. Use generating functions to find an explicit formula for the Fibonacci numbers.

3. Let H_n be the minimum number of moves needed to shift n rings from Peg 1 to Peg 2. One is not allowed to place a larger ring on top of a smaller ring. Find a recurrence relation for H_n and solve it using generating functions.



4. A girl has Rs. 100. Everyday she buys one of a chocolate (of Rs 1), an Ice-Cream (of Rs 5) or a Pastry (of Rs 5). How many ways are there (sequences) for her to spend Rs 100? Use generating functions.
   Ex. CPIIPCI represents "Day 1, buy Chocolate. Day 2, buy Pastry, day 3 Ice-cream etc.".

5. Use generating functions to solve the recurrence relation $a_k = 2a_{k-1} + 3a_{k-2} + 4^k + 6$ with initial conditions $a_0 = 20, a_1 = 60$.

6. Use generating functions to find the number of ways to make change for Rs 100 using notes of Rs 10, 20, 50.

7. Find a recurrence relation and generating function for the number of ways that the sum n can be obtained when a die is rolled repeatedly and the order of the rolls matters.

8. If $G(x)$ is the generating function for the sequence $\{a_k\}_0^\infty$, what is the generating function for the sequence: $0, 0, 0, a_3, a_4, a_5, ...$?

9. Exponential generating function for the sequence $\{a_n\}$ is the series: $\sum_{i=0}^{\infty} \frac{a_n}{n!} x^n$. Note that $e^x = \sum_{i=0}^{\infty} \frac{1}{n!} x^n$.
   Find a closed form for the exponential generating function for the sequence $\{a_n\}$, where a) $a_n = n$; b) $a_n = n(n-1)$

Indian Institute of Information Technology-Vadodara
MS441: Discrete Mathematics
Tutorial 11

1. Suppose that $A = \{1, 2, 3, 4\}$ and R be the relation on A defined as $(a, b) \in R$ iff $a < b$. Find the matrix, graph representation of R with respect to the natural ordering.

2. Suppose that the relation R on a set is represented by the matrix

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

   Is R reflexive, symmetric, antisymmetric?

3. Determine whether the relation R on the set of real numbers- $\mathbb{R}$ is reflexive, symmetric, antisymmetric, transitive, where $(x, y) \in R$ if and only if
   (i) $x + y = 0$
   (ii) $xy \geq 0$
   (iii) $x = 1$ or $y = 1$

4. Let A be the relation "to be wife of" and B "to be father of " on the set of all humans. What does the relation $A \circ B$ mean in this case?

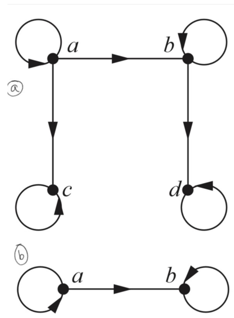
5. Let $A = \{2, 3, 4, ..., 100\}$ with partial order of divisibility.
   (a) How many maximal elements does $(A, |)$ have?
   (b) Give a subset of A that is a linear order under divisibility and is as large as possible.

6. A person's blood type is determined by the presence (T) or absence (F) of antigens A, B and Rh, as shown in the table below.

A	B	Rh	Type
F	F	F	O^-
F	F	T	O^+
F	T	F	B^-
F	T	T	B^+
T	F	F	A^-
T	F	T	A^+
T	T	F	AB^-
T	T	T	AB^+

   A person with blood type X can donate blood to a person with blood type Y, if and only if all of the antigens present in X are contained in Y. Let P be the set of the eight possible blood types, and let R be the relation on P such that XRY if and only if a person with blood type X can donate blood to a person with blood type Y. Answer the following questions.
   (a) Can a person with A^+ blood type donate to one with A^- ?
   (b) What types of blood can a person with A^+ blood type receive?
   (c) Draw a directed graph for R.
   (d) Show that R is a partial order.
   (e) Make a Hasse diagram for R.
   (f) What are the minimal (universal donor) and maximal (universal acceptor) elements of P?

7. Determine whether the relation with the directed graph shown is a partial order.



8. Display all the partial orders on a set with three elements with the help of Hasse diagram. How many of them are lattices?

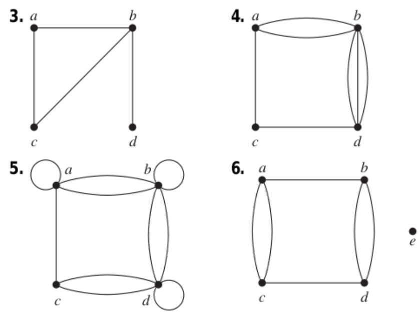
9. Let R be a partial order on a finite set S. Describe how to use the matrix representation M_R to find the least and greatest element of A if they exist.
Greatest element: $y \in (S, \preceq)$ is greatest if $x \preceq y$ <u>for all</u> $x \in S$.
Least element: $z \in (S, \preceq)$ is least if $z \preceq x$ <u>for all</u> $x \in S$.

Indian Institute of Information Technology-Vadodara

MS441: Discrete Mathematics

Assignment

1. Determine whether the graphs shown below has directed or undirected edges, whether it has multiple edges, and whether it has one or more loops.



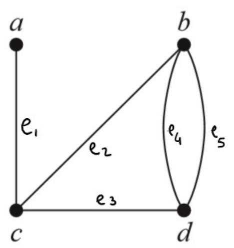
2. Suppose that there are four employees in the computer support group of the School of Engineering of a large university. Each employee will be assigned to support one of four different areas: hardware, software, networking, and wireless. Suppose that Ping is qualified to support hardware, networking, and wireless; Quiggley is qualified to support software and networking; Ruiz is qualified to support networking and wireless, and Sitea is qualified to support hardware and software.
Use a bipartite graph to model the four employees and their qualifications. Does there exists maximal/complete matching?

3. Construct the graph for a set of seven telephone numbers 555-0011, 555-1221, 555-1333, 555-8888, 555-2222, 555-0091, and 555-1200 if there were three calls from 555-0011 to 555-8888 and two calls from 555-8888 to 555-0011, two calls from 555-2222 to 555-0091, two calls from 555-1221 to each of the other numbers, and one call from 555-1333 to each of 555-0011, 555-1221, and 555-1200.

4. Can a simple graph exist with 15 vertices each of degree five?

5. How many edges does a graph have if its degree sequence is 4, 3, 3, 2, 2? Draw such a graph. Is it bipartite?

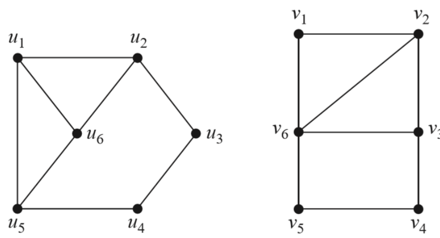
6. Represent following graph using incidence matrix and adjacency matrix



7. Draw an undirected graph represented by following adjacency matrix:

$$\begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

8. Are following graphs isomorphic?



9. Show that in a simple graph with at least two vertices there must be two vertices that have the same degree.

10. How many subgraphs with at least one vertex does K_3 have?

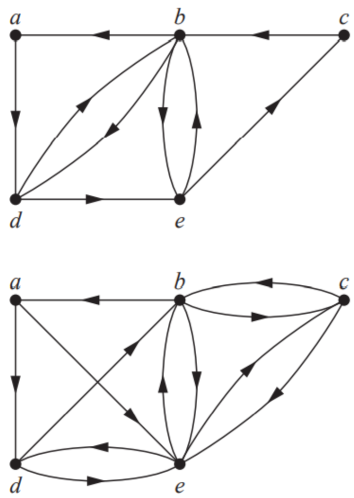
11. Let G be a graph with v vertices and e edges. Let M be the maximum degree of the vertices of G, and let m be the minimum degree of the vertices of G. Show that $m \leq 2e/v \leq M$.

12. Suppose that a connected planar simple graph has 20 vertices, each of degree 3. Into how many regions does a representation of this planar graph split the plane?

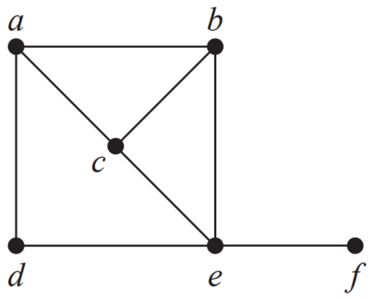
13. Can five houses be connected to two utilities without connections crossing?

14. What are the chromatic numbers of K_n, C_n, W_n?

15. Does following directed graphs have Euler circuit? If yes then find it else give justification. Find conditions for existence of Euler circuit in a directed graph with no isolated vertex.

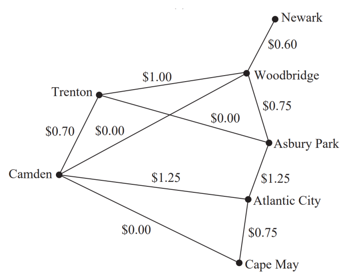


16. Determine whether the given graph has a Hamilton circuit or path.

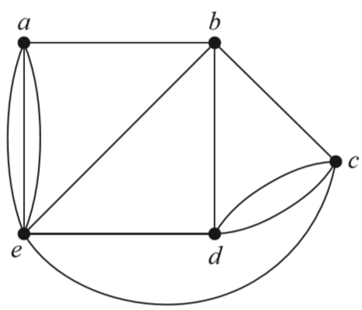


17. For which values of m and n does the complete bipartite graph $K_{m,n}$ have an a) Euler circuit? b) Euler path?

18. Find a shortest route in distance between Newark and Camden, and between Newark and Cape May, using Dijkstra's algorithm. Write down each iteration while implementing the algorithm



19. Can you draw following graph without lifting your pen and without tracing an edge twice? If yes then draw it. Did you stop at a vertex where your started it?



20. An edge coloring of a graph is an assignment of colors to edges so that edges incident with a common vertex are assigned different colors. The edge chromatic number of a graph is the smallest number of colors that can be used in an edge coloring of the graph. What is the edge chromatic number of the graph given in previous question?

21. Represent following map of Gujarat using a graph. How many minimum colours are required to paint it so that adjacent districts have different colours?


Banaskantha
Kutch
Patan
Sabar Kantha
Mehesana
Arvalli
Gandhinagar
Mahisagar
Ahmedabad
Kheda
Panch Mahal
Dahod
Morbi
Surendra Nagar
Anand
Jamnagar
Vadodara
Udepur
Dwarka
Botad
Rajkot
Narmada
Bharuch
Porbandar
Bhavnagar
Junagadh
Amreli
Surat
Tapi
Somnath
Dang
Navsari
Valsad